



**Verified Carbon
Standard**

SINOP LANDFILL GAS TO ELECTRICITY PROJECT

Document Prepared by Life İklim ve Enerji Ltd. Şti.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Sinop Landfill Gas to Electricity Project (hereafter referred to as the project) is located at Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey. The project is owned by Biokütle Enerji Elektrik Üretim A.Ş. (hereafter referred to as the project owner). The project is designed to utilize the gas produced in the existing landfill to generate electricity. The provisional acceptance of the facility is 29-June-2021¹ with half capacity of 1.453 MWm / 1.414 MWe with single generator, but it started its operation as of 01-July-2021. The total installed capacity will be 2.906 MWm / 2.828 MWe (2x1.453) MWm / (2x1.414) MWe, consisting of 2 sets of generators, and installation of second generator is planned to be completed as of 2024. The feed-in electricity is estimated to be 10,605 MWh/year with half capacity and 21,210 MWh / year² with full capacity. It will handle an average of 174³ tons MSW per day.

Prior to the implementation of the project, the landfill site did not have gas extraction and utilization systems installed, and the LPG was directly emitted to the atmosphere. The project collects and utilizes LFG generated by Sinop landfill for power generation. Therefore, the project drastically reduces methane emissions, and at the same time, electricity is generated from the collected gas. Electricity generated is provided to the grid and the facility utilizes electricity generated by the project instead of drawing electricity from the grid.

The environmental, socio-economic, and technological benefits of the Project are described as follows:

- Reduction in fossil fuel use (imported or local) by using renewable energy resources,
- Reduction in greenhouse gas emissions from the landfill area by using biogas for electricity production,
- Reduction in energy production costs and imported energy amounts,
- Improvement of environmental conditions (GHG and odor) and safety in the landfill area.

The project is expected to reduce 40,803 tCO₂e/year in average in the first 7-year crediting period between commission date of 01-July-2021 and ending date of 30-June-2028. Vintage reductions are given in the table below:

¹ Sinop Landfill Provisional Acceptance

² EMRA Generation License

³ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>WVB Name</u>	<u>Number of years</u>
Joint Validation & Verification	01-July-2021 – 28-February 2023	VCS	Applus	1 years and 8 months
<u>Total</u>	01-July-2021 – 28-February 2023			1 years and 8 months

1.2 Sectoral Scope and Project Type

Scope 1: energy industries (renewable-/non-renewable sources)

Scope 13: Waste handling and disposal.

Methodology:

AMS-III.G. Landfill methane recovery.

AMS-I.D Grid-connected renewable electricity generation

The project is not a grouped project.

1.3 Project Eligibility

The project is an LFG power generation project, utilizing landfill gas (LFG), which consists mainly of methane, for electricity generation, which is eligible under the scope of the VCS Program.

1.4 Project Design

The project includes a single location or installation only

Eligibility Criteria

The project is not a grouped project, so it is not applicable.

1.5 Project Proponent

Organization name	Biokütle Enerji Elektrik Üretim Anonim Şirketi
Contact person	Savaş Uzun
Title	CEO

Address	Soğuksu Mah. 308 Sok. Adabel Sit. No:42/5 Muratpaşa, Antalya
Telephone	+90 (242) 237 31 00
Email	savas.uzun@biosenerji.com

1.6 Other Entities Involved in the Project

Organization name	Life İklim ve Enerji Ltd. Şti.
Role in the project	Carbon Certification Consultant
Contact person	Ramazan Aslan
Title	Manager
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara
Telephone	+90 312 481 2142
Email	ramazan.aslan@lifeenerji.com

1.7 Ownership

Biokütle Enerji Elektrik Üretim Anonim Şirketi holds all licenses, permits, and land use rights related to Sinop LFG and has obtained and owns all equipment used for its implementation.

1.8 Project Start Date

As it is stated in the Sinop Landfill Provisional Acceptance the start date of the project is 29-June-2021⁴. However, project started to utilize methane and generate electricity as of 01-July-2021 which can be seen in EPIAS records. Therefore, start date of the project is 01-July-2021 as per VCS standard 4.4 paragraph 3.8.

1.9 Project Crediting Period

Crediting Period Type: Renewable (Renewable twice for a total of 21 years)

Crediting Period Start date: 01-July-2021

Crediting Period End date: 30-June-2028

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

⁴ Sinop Landfill Provisional Acceptance

The estimated annual GHG emission reductions/removals of the project are 40,803 tCO₂e /annum. Same is below 300,000 tCO₂e.

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December-2021	12,979
01-January-2022 – 31-December-2022	29,743
01-January-2023 – 31-December-2023	33,186
01-January-2024 – 31-December-2024	42,173
01-January-2025 – 31-December-2025	44,917
01-January-2026 – 31-December-2026	47,364
01-January-2027 – 31-December-2027	49,562
01-January-2028 – 30-June-2028	25,703
Total estimated ERs	285,627
Total number of crediting years	7
Average annual ERs	40,803

1.11 Description of the Project Activity

The Project, which is located in the Merkez District of Sinop, utilizes the landfill gas (LFG) generated by the existing landfill for energy generation. The Project has an operation period of 20 years under the current Project Owner. The Project Owner will then transfer the rights of operation of the Project to the Sinop Municipality at the end of the period.

Waste projections conducted under feasibility studies⁵ indicate that the landfill receives 173.6 t/day waste on average and the waste quantity is expected to rise during operation period from 155 t/day in 2021 to 236 t/day in 2040.

⁵ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

Waste is collected and deposited to the landfill by the Municipality. Then, landfill gas is extracted via landfill gas collection system. Collected gas is flared and utilized for electricity generation.

Steps in the landfill gas to electricity project:

- 1- Storage of waste in landfill
- 2- Generation of LFG at the landfill
- 3- Collection of LFG via pipelines
- 4- Booster system that draws collected LFG
- 5- Treatment of LFG via Chiller Unit
- 6- Electricity and thermal energy generation in gas engines
- 7- Electricity feeding to national power grid
- 8- Retaining extra gas, that is not burnt in the engines, in the Gas holder

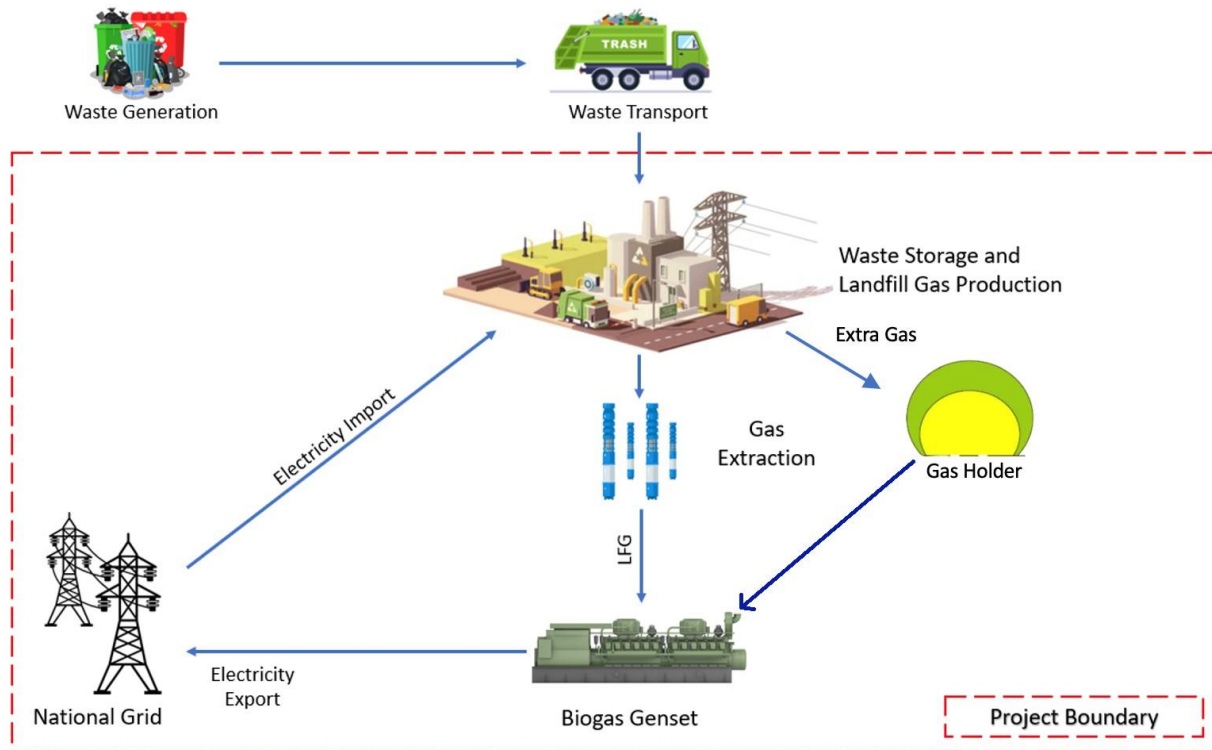


Figure 1 Boundary and the process flow of the Project

The collection system which consists of pipelines and booster systems extracts gas from landfill and transmits LFG to chiller unit.

Chiller unit treats LFG and removes impurities and moisture from the gas. LFG is then transmitted to gensets.



Figure 2 Ürker Chiller UC ASH 15.1⁶

Table 1 Chiller technical specifications

Brand	Ürker Chillers
Model	UC ASH 15.1
Compressor Type	Semi-hermetic
Production year	2021
Chilling capacity	37.24 kW/h
Total power drawn	13.58 kW/h

The gas holder is designed to securely store excess LFG gas using durable, corrosion-resistant materials and reliable sealing mechanisms. It features multiple layers, including an inner membrane and an outer protective layer, ensuring safety and integrity. The holder's capacity is based on LFG gas production and consumption rates, preventing overpressure and hazards during genset shutdowns. After the LFG is retained in the gas holder, it is utilized to produce electricity once the genset is operational again. This system ensures a continuous LFG gas supply, accommodating production fluctuations without leaks or emissions.

Table 2 Gas Holder technical specifications

Brand	Ecomembrane
Model	2MASTER®
External Shape	$\frac{3}{4}$ sphere
Production year	2021

⁶ <https://www.urkerchillers.com/tr/img/pdf/Hava-Sogutmalı-Chiller-Katalogu.pdf>

Volume	3500 m ³
--------	---------------------

Gensets generates electricity which is then fed to the grid.

The Project will utilize 2 units of Jenbacher brand gensets with a capacity of 1.414 MWe each. Currently only one unit is in operation and the second unit is planned to start operation in 2024⁷. The Project will have the capacity to generate 2.828 MWe when it reaches full capacity.

Technical Description

Genset
JGS 420 GS-L.L

static Grid Code

Sinop LF-2020-1



Electrical output 1414 kW el.

Emission values
NOx < 500 mg/Nm³ (5% O₂) | < 190 mg/Nm³ (15% O₂)

Figure 3 Jenbacher Generator Technical Specifications

⁷ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

0.01 Technical Data (at genset)

			100%	75%	50%
Power input	[2]	kW	3.403	2.625	1.848
Gas volume	*)	Nm³/h	851	656	462
Mechanical output	[1]	kW	1.453	1.089	727
Electrical output	[4]	kW el.	1.414	1.061	706
Heat to be dissipated (calculated with Glykol 37%)					
~ Intercooler 1st stage (Engine jacket water cooling circuit)	[9]	kW	291		
~ Intercooler 2nd stage (Low temperature circuit)		kW	75		
~ Lube oil (Engine jacket water cooling circuit)		kW	201		
~ Jacket water		kW	383		
~ Surface heat	ca. [7]	kW	122		
Spec. fuel consumption of engine electric	[2]	kWh/kWel.h	2,41	2,48	2,62
Spec. fuel consumption of engine	[2]	kWh/kWh	2,34	2,41	2,54
Lube oil consumption	ca. [3]	kg/h	0,29	~	~
Electrical efficiency	up to 35°C T1		41,5%	40,4%	38,2%

Figure 4 Jenbacher Generator Technical Specifications⁸

The total annual power generation is estimated to be 21,210 MWh/year. The expected average annual emission reductions are 40,803 tCO₂e in the first 7-year crediting period. The total estimated emission reductions over the crediting period of the project will be 285,627 tCO₂e. The milestone table of the project is presented below.

1.12 Project Location

The project is in Hacıoglu Neighborhood, which is located in Merkez District of Sinop Province in Turkey. The GPS coordinates of the Project area are provided in the following table.

Table 3 Project Area GPS coordinates

N	E
41.949269823	35.036668530
41.949272309	35.036911608
41.949194300	35.036913527
41.949194861	35.036979850
41.948934092	35.036983778
41.948930761	35.036673646

⁸ <https://www.innio.com/en/jenbacher/gas-engines/type-4/j420>



Figure 5 Location of the Project



Figure 6 Google Earth view of the Project and Landfill Site

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the Project is the same as the baseline scenario described in Section 3.4.

- Prior to the implementation of the project, the landfill site did not have gas extraction and utilization systems installed.
- Gas generated due to decaying organic matter was released directly to the atmosphere.
- Equivalent electricity generated by the project was supplied by the grid, which is dominated by fossil fuel-based power plants.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project owner acquired all necessary legal permits prior to the commission of the facility. Most important legal documents are listed.

Project owner signed a contract⁹ with Union of Sinop Coastal Municipalities for operation of landfill and electricity generation from landfill gas on 11-November-2020.

The project has received an Environment Impact Assessment Exemption¹⁰ from Sinop Provincial Directorate of Environment and Urbanization on 16-June-2021.

Project has received the Electricity Generation License (EGL) on 24-June-2021¹¹.

Legal Framework

Laws and regulation concerning electricity generation and landfill operation are listed below. The regulations on waste management require precautions to prevent the explosion of landfill gas but do not require recovery or destruction.

The Project complies with all applicable laws and regulations.

Related Laws

- Electricity Market Law¹² (Law Number: 6446, ratified 14-March-2013, enacted 30-March-2013)

The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market.

⁹ Contract For the Operation of Landfill Belonging to The Union of Sinop Coastal Municipalities and the Generation of Electrical Energy from The Landfill Gas

¹⁰ Environment Impact Assessment Exemption Letter

¹¹ Electricity Generation License

¹² <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf>

No regulatory requirement for destruction of landfill gas.

- Law on Utilization of Renewable Energy Resources for the Purpose of Generating Electricity Energy¹³ (Law Number: 5346, ratified 10-May-2005, enacted 11-August-1983)

The purpose of this Law is to expand the utilization of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the environment and to develop the related manufacturing sector for realizing these objectives.

The Law brings an incentive of 13.3 \$ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration

¹³ <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf>

- Environment Law¹⁴ (Law Number: 2872, ratified 09-August-1983, enacted 11-August-1983)

This Law addresses the ecological security of the population, the rational use of natural resources, nature conservation and environmental protection. Additional Article 6 says that clean air policies should be applied in provinces and districts, and air quality should be monitored. Methodologies for determination, monitoring and measurement of air quality, air quality limit values, precautions to prevent air pollution and public awareness are responsibilities of the Ministry of Environment and Forestry.

No regulatory requirement for destruction of landfill gas.

Related Regulations

- Regulation on Solid Waste Management¹⁵ dated 02-April-2015 and numbered 29314

The regulation aims to determine general waste management principles to protect the environment and human health from generation to disposal of waste.

No regulatory requirement for destruction of landfill gas.

- “Regulation on landfilling¹⁶” dated 26-March-2010 and numbered 27533

The regulation aims to protect the environment by minimizing the negative impacts of leachate and landfill gas on soil, air, underground and surface water

No regulatory requirement destruction of landfill gas.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has not been registered or is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

Since the Project did not apply for any other programs, it did not get rejection from any either.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

¹⁴ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5>

¹⁵ https://webdosya.csb.gov.tr/db/cygm/editordosya/Atik_Yonetimi_Yonetmeligi-2017.docx

¹⁶ <http://www.mevzuat.gov.tr/Metin.Aspx?MevzuatKod=7.5.13887&MevzuatIliski=0&sourceXmlSearch=>

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

Not applicable. The project activity is neither included in an emissions trading program nor does it take place in a jurisdiction or sector in which has binding limits on GHG emissions.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The project has not sought or received another form of GHG-related environmental credit or renewable energy certificates.

1.16.3 Supply Chain (Scope 3) Emissions

The project will remove methane emissions by capturing and destroying LFG and will remove carbon by electricity supplying to the national grid. Apart from that project activity will not impact on any scope 3 emissions.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project not only aims to generate energy from a sustainable and renewable energy source but also to find a sustainable solution to an unmanaged landfill which released landfill gas to atmosphere prior to the installation of facility. The Project will contribute to long term sustainable development of the region.

Main contributions to the Sustainable Development Goals¹⁷ adopted by the United Nations are summarized.

SDG 7: Affordable and Clean Energy

- The project converted previously unmanaged landfill into a biogas-to-energy facility which is also a clean and efficient facility for waste disposal and handling.
- Generates energy from a renewable source.

¹⁷ <https://www.undp.org/sustainable-development-goals>

- Provides energy to the grid from renewable a source and replaces energy generated by fossil fuel utilization.

SDG 8: Decent Work and Economic Growth

- Creates employment opportunities related to the management, operation and maintenance of landfill and the methane recovery and electricity generation systems.
- Turkey is an energy and fuel dependent country, and the project contributes to reduction of energy dependency for the country and thus contributes to economic growth.

SDG 13: Climate Action

- Management of unmanaged landfill site by establishing a new, clean and efficient facility for waste disposal and handling.
- Collection of landfill gas, reduction and prevention of direct emission of GHG to the atmosphere. Estimations indicate that The Project can reach an average emission reduction of 30,789 tCO₂eq annually.

1.17.2 Sustainable Development Contributions Activity Monitoring

The project, located in Sinop Turkey, reduces GHG emissions by capturing and utilizing methane generated in the landfill. Project prevents direct methane emissions from the landfill to the atmosphere via collection system and generates renewable energy via flaring methane and by doing so increases renewable energy share in the Turkish electricity grid. Reduces amount of wastewater generated by displacing energy generated by fossil fuel plants that use water for cooling. The project also creates equal and safe employment opportunities in the region.

Table 4: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	Implemented activities to increase	The Project Activity will provide employment to at least 9 people.	Project Proponent provides employment to at least 9 people during the project lifetime complying with Labor Law in Turkey.
2)	8.8	Protect labour rights and promote safe working environments	Implemented activities to increase	Occupational Health and Safety training is provided to employees at least once a year	Occupational Health and Safety training is provided to employees at least once a year
3)	7.2	By 2030, increase substantially the share of renewable energy in the global energy mix	Implemented activities to increase	Total electricity generation during the monitoring period is 8,123 MWh.	Total electricity generation for life time of the project is 8,123 MWh.

4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Total emission reduction during the monitoring period is 41,132 tCO ₂ e.	Total emission reduction for life time of the project is 41,132 tCO ₂ e.
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1.18 Additional Information Relevant to the Project

Leakage Management

In accordance with the methodologies applied to the project, no leakage effects are accounted for.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

N/A

2 SAFEGUARDS

2.1 No Net Harm

Project owner applied for an evaluation regarding Environmental Impact Assessment (EIA) Regulation to Sinop Provincial Directorate of Environment and Urbanization on 23-December-2020 and the Project received EIA Exemption after evaluation of the project from the Directorate on 16-June-2021.

The old practice on the site was a landfill with no methane recovery. Now, landfill gas is utilized to generate power with the project activity. The Project has several environmental benefits:

- It will cover the landfill area with soil, reducing the adverse visual impacts.
- The cover and an effective landfill gas collection system will reduce odour from the landfill
- The LFG recovery will reduce GHG emissions and help in tackling global warming
- It will set a good example for proper waste management with improved health and safety impacts
- It will minimize explosion and fire risks by decreasing gas accumulation
- It will feed clean energy to the grid, thereby reducing emissions from power generation

As one of the sustainable renewable energy generation technologies, the Project reduces the environmental impacts of the current practice on the landfill site considerably.

2.2 Local Stakeholder Consultation

The project's local stakeholder consultation meeting took place on 18-October-2022 in Sinop Landfill, operation building. The local stakeholders were informed about the stakeholder meeting via announcements which were posted on 05-October-2022. The announcements were placed around high populated areas of the neighborhood to be seen by the locals.

The company invited all potential stakeholders to participate in the local stakeholder consultation meeting to brief them about the project and gather their feedback regarding the implementation of the project and VCS application of the project. In addition, potential stakeholders such as NGOs, ministries, governments etc. were invited to the meeting through 06 October 2022 dated e-mail¹⁸.

Some local residents from Hacıoglu Village and a member of Union of Municipalities of Sinop has attended to the meeting.

During the meeting a presentation was provided detailing the project, its grievance mechanisms, environmental and social impacts, contributions to sustainable development goals.

In addition, during the local stakeholder consultation meeting, participants filled in an evaluation form regarding the project to document their feedback on the impacts of the project. Attendance is recorded.

In general, local stakeholders were supportive of the project activity. Most of the local stakeholders thought the Project would help improve the livelihood of local people and promote local economic development without much adverse environmental impact. Therefore, the project implementation is regarded as beneficial by most of the stakeholders. During the meeting, the stakeholders did not raise any concerns about the project, and they indicated since the implementation of the project waste and odor issue caused by the wild dumpsites disappeared.

A logbook for grievance mechanism has delivered to the Muhtar and grievance mechanism was established with the support of the Mukhtar for stakeholders to forward their requests or concerns regarding the project activities. In Turkey, mukhtars are the most relevant persons in the villages for establishing a grievance mechanism. The Mukhtar was provided all the contact information of the responsible persons from the project activity. Whenever a complaint arises from the local people or Mukhtar himself arises a concern regarding the project activity, he can directly contact the relevant responsible persons.

2.3 Environmental Impact

The environmental matters in Turkey are jurisdiction of the Ministry of Environment and Urbanization. The Law defines among others the general requirements of the Environmental

¹⁸ Announcement and sent e-mail have been provided.

Impact Assessment (EIA) states specific requirements according to the type of project, location, size, and other characteristics.

Project owner applied for an evaluation regarding Environmental Impact Assessment (EIA) Regulation to Sinop Provincial Directorate of Environment and Urbanization on 23-December-2020 and the Project received EIA Exemption after evaluation of the project from the Directorate on 16-June-2021.

The landfill area is operated in accordance with “Solid Waste Facilities Field Management and Operating Guideline” published by the Ministry of Environment and Urbanization. In accordance with the guideline, the landfill is covered by soil layer with height of 0.3m -0.5m to reduce environmental impact of landfill site.

2.4 Public Comments

Following initial project listing, the VCS/Verra project Sinop Landfill Gas to Electricity Project was open for 30 days of public consultation from 12-January 2023 to 11-February-2023. No comment has been received during public consultation.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Approved methodologies applied to the project are;

1. AMS-III.G. Landfill methane recovery Version 10.0

This methodology is described as “*Capture and combustion of methane from landfills used for disposal of residues from human activities including municipal, industrial and other solid wastes containing biodegradable organic matter*”¹⁹ in the CDM Methodology Booklet published by United Nations.

AMS-III.G Version 10 refers to following tools where applicable:

- TOOL04 - “Emissions from solid waste disposal sites” version 08.1
- TOOL 05 - “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” version 03.0
- TOOL03 - “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” Version 03.0

¹⁹ https://cdm.unfccc.int/methodologies/documentation/meth_booklet.pdf

- TOOL09 - “Determining the baseline efficiency of thermal or electric energy generation systems” Version 03.0
- TOOL32 - “Positive lists of technologies” Version 04.0

2. AMS-I.D Grid-connected renewable electricity generation Version 18.0

AMS-I.D Version 18 refers to following tools where applicable:

- TOOL07 - "Tool to calculate the emission factor for an electricity system" version 07.0
- TOOL03 - “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” Version 03.0
- TOOL10 - “Tool to determine the remaining lifetime of equipment” Version 01
- TOOL16 – “Project and leakage emissions from biomass” Version 05.0
- TOOL11 - “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1

3.2 Applicability of Methodology

The baseline calculation for the Project follows the procedures for categories:

- Small-scale Methodology: Landfill methane recovery AMS-III.G / Version 10;

This methodology applies to Sinop landfill gas project activity, where the baseline scenario is: in the absence of the project activity, MSW is left to decay within the project boundary, and methane is emitted into the atmosphere. There was no legal requirement for capturing the methane emissions in the baseline scenario. Furthermore, for this project activity, it is estimated that annual emission reductions will be less than 60 kt CO₂e which is lower than the threshold value specified in the applied methodology. Also, in the absence of the project activity, organic waste would not be recycled and just left to decay. Moreover, in this project activity, the management of the solid waste disposal site wasn't deliberately changed to increase methane generation compared to the situation before the implementation of the project activity.

Applicability of Methodology	Justification
AMS-III.G., “Landfill methane recovery” version 10.0	
Different options to utilise the recovered landfill gas as detailed in paragraph 4 of “AMS-III.H.: Methane recovery in wastewater treatment” (version 19.0) are eligible for use under this methodology. The relevant	Landfill gas is utilized for electrical energy generation directly, so it is eligible and applicable.

procedures in AMS-III.H. shall be followed in this regard.	
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	The Project Activity is estimated to result in less than 60 kt CO ₂ equivalent annually, and 30 kt CO ₂ average at max capacity. So, this requirement applies to the Sinop LFG Project.
The proposed project activity does not reduce the quantity of organic waste that would have been recycled in the absence of the project activity.	Sinop LFG Project activity does not reduce the quantity of organic waste that would have been recycled in the absence of the project activity.
The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity.	There was no organic waste recycling prior to project, so project activity does not reduce amount of organic waste that would have been recycled.
This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement).	The management of the landfill site belongs to the Municipalities 'Union so that Project Proponent does not have any control over the SWDS management. Thus, the SWDS management is not deliberately worsened to increase methane generation prior to the project implementation.

The applicability conditions of the tools referred by the applied methodology AMS-III.G. are explained in the following table.

Applicability of Methodology	Justification
AMS-III.G., "Landfill methane recovery" version 10.0	
"Emissions from solid waste disposal sites" version 08.0	The Project Activity falls under Application A. project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring the methane. The tool is only applied for an ex-ante estimation of emissions in this report. The emissions will then be monitored during the crediting period.
"Tool to calculate project or leakage CO2 emissions from fossil fuel combustion" Version 03.0	Not applicable. Emissions referred to are negligible compared to the project emissions from electricity consumption or flaring.
"Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" Version 03.0"	This tool is applied to calculate the project emissions from electricity consumption by the project activity. Scenario, A of Paragraph 5 (a) of the tool, apply for this project activity. Electricity consumption is from the grid, and there are no captive power plants on site.
"Determining the baseline efficiency of thermal or electric energy generation systems" Version 03.0	Not applicable. Turkish Ministry of Energy and Natural Resources publishes Emission factor applicable for the national grid for wind, solar and other renewable projects.
"Positive lists of technologies" Version 03.0	This tool is applied for the demonstration of the additionality of the Project Activity. According to Paragraph 5 of the tool, the project activity falls under the category of 5.1.1 Landfill gas recovery and its gainful use. The LFG is used to generate electricity in a power plant with a total capacity below 10MW.
Applicability of Methodology	Justification
AMS-I.D., "Grid-connected renewable electricity generation" version 18.0	
This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s);	The Project is a Greenfield project involving the setting up of a new power plant at a site where there were no power plants in the pre-project scenario. The Project is only 2.828 MWe. Therefore, the project is applicable under (a) clause.

(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m² ; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m²	The project is not a hydro power plant, so the criteria is not applicable.
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The Project Activity doesn't have any non-renewable component. The Project involves electricity generation by utilizing LFG produced from a landfill with a total capacity of 2.828 MWe.
Combined heat and power (cogeneration) systems are not eligible under this category.	The Project Activity is not a Combined heat and power (cogeneration) system. The Project's recovered methane is used only for electricity generation and supplied to the grid. It is not utilized for heat generation or cogeneration.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation	The Project Activity does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility

facility, the added capacity of the units added by the Project should be lower than 15 MW and should be physically distinct from the existing units.	since there was no renewable energy generation in the project site prior to Sinop LFG.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated, or replacement power plant/unit shall not exceed the limit of 15 MW.	The Project Activity is not a retrofit, rehabilitation or replacement project.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration, other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The recovered methane is used for electricity generation and supplied to the grid. The baseline for the electricity component is in accordance with the procedure prescribed under this methodology. The recovered methane is not used for heat generation or cogeneration.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The methane is not sourced from dedicated plantations.

The applicability conditions of the tools referred by the applied methodology **AMS-I.D.** are explained in the following table.

Applicability of Methodology	Justification
AMS-I.D., "Grid-connected renewable electricity generation" version 18.0	
"Tool to calculate the emission factor for an electricity system" version 07.0	Not applicable. Turkish Ministry of Energy and Natural Resources publishes Emission factor applicable for the national grid for wind, solar and other renewable projects.

“Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” Version 03.0	Not applicable. Emissions referred to are negligible compared to the project emissions from electricity consumption or flaring.
“Tool to determine the remaining lifetime of equipment” Version 01	This tool is not used for the determination of equipment lifetime. The Project is designed for an operational lifetime of 20 years, according to the contract with the Solid Waste Management Union of Sinop Coastal Municipalities.
“Project and leakage emissions from biomass” Version 04.0	Not applicable. Project Activity doesn't include biomass.
“Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1	Not applicable. Project Activity is not at the renewal of the crediting period.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for since the CO ₂ is also released under the project activity
		CH ₄	Yes	Major source of emissions in the baseline.
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from SWDS. This is conservative
		Other	No	Excluded for simplification. This is conservative
	Emissions from electricity generation	CO ₂	Yes	Major emission source if power generation is included in the project activity
		CH ₄	No	Excluded for simplification. This is conservative
		N ₂ O	No	Excluded for simplification. This is conservative
		Other		
Project		CO ₂	Yes	May be an important emission source.

Source	Gas	Included?	Justification/Explanation
Emissions from electricity consumption due to the Project activity	CH ₄	No	excluded for simplification. This emission source is assumed to be very small
	N ₂ O	No	excluded for simplification. This emission source is assumed to be very small
	Other	No	Excluded for simplification. This is conservative
	CO ₂	No	It is part of the natural carbon cycle, where the carbon was previously sequestered from the atmosphere.
	CH ₄	No	Excluded for simplification. This is conservative.
	N ₂ O	No	Excluded for simplification. This is conservative.
LFG capture and flaring	CO ₂	No	It is part of the natural carbon cycle, where the carbon was previously sequestered from the atmosphere.
	CH ₄	No	Excluded for simplification. This is conservative.
	N ₂ O	No	Excluded for simplification. This is conservative.
	Other	No	Excluded for simplification. This is conservative.

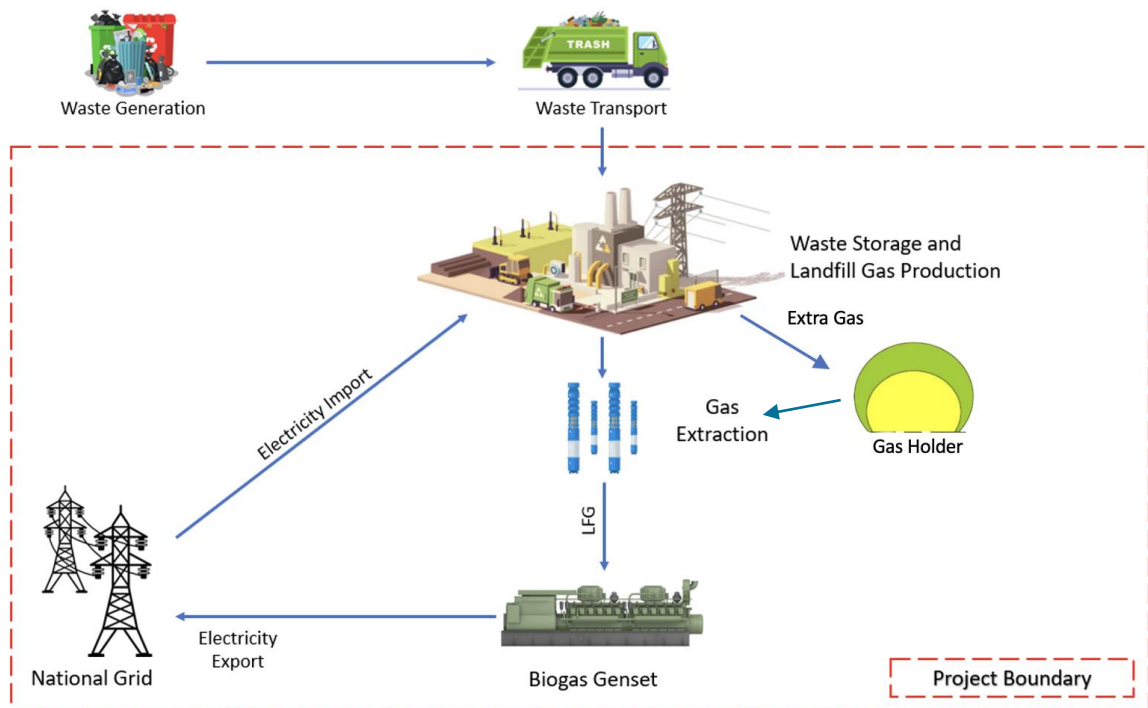


Figure 7 Boundary of the Project

3.4 Baseline Scenario

In the baseline scenario the biomass and organic matter decays in the landfill and methane is produced during the process. The produced methane is then released directly to the atmosphere without the proposed project activity. This scenario is identical to the baseline scenario described in **AMS-III.G**. There is no incentive or a national regulation/standard for prevention of methane emissions originating from a landfill. Baseline emission calculations covers the effects of methane oxidation difference of with and without project scenario.

In the absence of the project, scenario also complies with the baseline scenario described in AMS-I.D. In this scenario the electricity is provided to the Turkish national grid from power plants that mainly utilizes fossil fuels.

In Turkey there is no legal framework for abatement of emissions of greenhouse gases from landfill. In the absence of specific laws and regulations regarding GHG emissions there is no incentive for capturing and utilization of the LFG.

Therefore, as explained above, baseline emissions for the Project are:

- A. Landfill Gas: In the absence of the project activity, biomass and other organic matter are left to decay within the project boundary and methane is emitted into the atmosphere
- B. Electricity: Product of electricity energy baseline expressed in kWh of electricity produced by the renewable generating unit multiplied by an emission factor. Combined margin (CM) is adopted for emission factor.

The Project owner does not have fossil fuel-fired power plants or control of emissions resulting from the baseline scenario. The Project partially reduces LFG emissions from landfill and replace the electricity generation from grid-connected fossil fuel-fired thermal power plants, thereby reducing greenhouse gas emissions.

3.5 Additionality

LFG capturing and destroying is not mandated by any law, statute or other regulations in Turkey.

The proposed Project is deemed additional, as defined by the applied approved methodologies.

AMS-III.G. Small Scale Methodology Landfill Methane Recovery (Version 10.0) states that project activities and PoAs at new and existing landfills are deemed additional automatically as long as in the baseline scenario LFG was only vented. Other criteria stated in the methodology also applies to the project; total electricity generation capacity has to be under 10MW, and the project capacity is 2.282 MW.

Therefore, the project is deemed automatically additional.

3.6 Methodology Deviations

Not applicable.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The project has started operation as of 01 July 2021. During the operation phase of the project there is no major event that impacts the GHG emission reductions occurred. Milestones of the project are given in the table below:

Table 5 Milestones of the project activity

Date	Action
December 2020	Feasibility Report
11-December-2020	Operation Contract Between Project Owner and Sinop Coastal Municipalities Union
16-June-2021	EIA Exemption Letter
24.June-2021	EMRA Generation License
25.-June-2021	Grid Connection Agreement
29-June- 2021	Sinop Landfill Provisional Acceptance
01-July-2021	Start of Operation and Crediting Period
18-October-2022	Local Stakeholder Consultation Meeting

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions calculations are determined according to AMS-III.G and AMS-I.D methodologies. These emissions can be divided into two parts. First part is from the methane emissions originating from the SWDS and the second part originates from the electricity generation using fossil fuels or supplied by the grid (electricity displacement) in the absence of the project activity.

$$BE_y = BE_{CH4,y} + BE_{electricity,y} \quad \text{(Equation-1)}$$

Where;

BE_y	=	Baseline emissions in year y (tCO _{2e} /y)
$BE_{CH4,y}$	=	Baseline emissions of methane from the SWDS in year y (tCO _{2e} /y)
$BE_{electricity,y}$	=	Baseline emissions associated with electricity displacement y (tCO _{2e} /y)

Baseline emission calculations according to AMS-III.G (version 10.0):

The ex-ante emission reduction by the project activity will be calculated according to AMS-III.G with the following equation:

$$BE_y = \eta_{PJ} \times BE_{CH4,SWDS,y} - (1 - OX) \times F_{CH4,BL,y} + GWP_{CH4} \quad \text{(Equation-2)}$$

Where;

$BE_{CH4,SWDS,y}$	=	Methane emission potential of a solid waste disposal site (tCO _{2e})
OX	=	Oxidation factor (A default value of 0.1 is used)
$F_{CH4,BL,y}$	=	Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (tCH ₄)
η_{PJ}	=	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50% is used
GWP_{CH4}	=	Global Warming Potential for methane

Determination of $F_{CH_4,BL,y}$

As per ACM0001 v19.0, 4 cases given in the table below has been evaluated.

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Case 1: No requirement to destroy methane exists and no existing LFG capture system

In this situation:

$$F_{CH_4,BL,y} = 0$$

Case 2: Requirement to destroy methane exists and no existing LFG capture system

$$F_{CH_4,BL,y} = F_{CH_4,BL,R,y}$$

$F_{CH_4,BL,R,y}$ should be determined based on the information contained in the requirement to destroy methane, as follows:

(a) If the requirement specifies the amount of methane that must be flared then that amount is $F_{CH_4,BL,R,y}$;

(b) If the requirement specifies a percentage of the captured LFG that is required to be flared, the amount shall be calculated as follows:

$$F_{CH_4,BL,R,y} = \rho_{reg,y} \times F_{CH_4,PJL,capt,y}$$

Where:

$F_{CH_4,BL,R,y}$ = Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (t CH₄/yr)

$\rho_{reg,y}$ = Fraction of LFG that is required to be flared due to a requirement in year y

$F_{CH_4,PJL,capt,y}$ = Amount of methane in the LFG which is captured in the project activity in year y (t CH₄/yr)

$F_{CH_4,PJL,capt,y}$ can be calculated by two options:

Option 1: Calculate using the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”, applying the following requirements:

- i. The gaseous stream tool shall be applied to the LFG pipeline immediately downstream of the LFG capture system and before any split in the gaseous flow to different uses or flares
- ii. CH₄ is the greenhouse gases for which the mass flow should be determined;
- iii. The simplification offered for calculating the molecular mass of the gaseous stream is valid (equations (3) or (17) in the tool); and
- iv. The mass flow should be calculated on an hourly basis for each hour h in year y.

Option 2: Calculate as the sum of the amount of methane that is sent to the flare, electricity generating or heat generating equipment in year y, however, not taking into account the working hours of the equipment;

- i. If the requirement does not specify the amount or percentage of LFG that should be destroyed but requires the installation of a capture system, without requiring the captured LFG to be flared then

$$F_{CH_4,BL,R,y} = 0$$

- ii. If the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare the LFG, then a typical destruction rate of 20 per cent is assumed:

$$F_{CH_4,BL,R,y} = 0.2 \times F_{CH_4,PJL,capt,y}$$

Case 3 and Case 4 is not evaluated since there is no existing LFG capture system in baseline and this situation will be valid always.

As a result;

$F_{CH_4,BL,y}$ is taken as zero since there is no regulation in Turkey that advises the capture of methane emissions and there is no existing LFG capture and destruction system. ($F_{CH_4,BL,y} = 0$), if regulation is changed Case 2 will be valid and used during verification of emission reductions.

$BE_{CH_4,SWDS,y}$ is calculated according to Tool 04 Emissions from solid waste disposal sites Version 08.0.

$$\left. \begin{array}{l} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{array} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \\ \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j})) \quad \text{(Equation-3)}$$

Where;

- $BE_{CH_4,SWDS,y}$ = Methane emission potential of a solid waste disposal site (tCO_{2e})
- x = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period ($x = 1$) to year y ($x = y$)

y	=	Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
$DOC_{i,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
$W_{j,x}$	=	Quantity of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
ϕ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
GWP_{CH_4}	=	Global Warming Potential for methane
OX	=	Oxidation factor (A default value of 0.1 is used)
F	=	Fraction of methane in the SWDS gas (volume fraction)
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction)
k	=	Decay rate for the waste type j (1 / yr)
j	=	Type of residual waste or types of waste in the MSW

The project gas collection efficiency is 85% according to the feasibility report.

Baseline emission calculations according to AMS.I.D (version 18.0):

According to AMS-I.D, baseline emissions include only CO₂ emissions from electricity generation in power plants displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_{y,electricity} = EG_{PJ,y} + EF_{grid,y} \quad \text{(Equation-4)}$$

Where;

$$BE_{electricity,y} = \text{Baseline emissions associated with electricity displacement } y \text{ (tCO}_{2e}/y)$$

EG_{PJ,y} = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

EF_{grid,y} = Combined margin CO₂ emission factor of grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

$$EG_{PI} = EG_{PI, facility, y} \quad \text{(Equation-5)}$$

Where;

EG_{PJ,facility,y} = Quantity of net electricity generation that is produced by the project plant to the grid in year y (MWh)

Determination of the emission factor for electricity generation ($EF_{grid,y}$)

To find $EF_{grid,y}$, “Tool to calculate the emission factor for an electricity system” (V 07.0) is applied to calculate combined margin. The following six steps below, provided in the Methodological Tool, are used to calculate the combined margin (CM) emission factor:

For the emission factors for electricity generation, the publication of the Turkish Ministry of Energy and Natural Resources, which indicates Turkey's National Electric Grid Emission Factor for 2022²⁰, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors, for the relevant year with the usage of the IPCC's Clean Development Methodology Tool 07-V06.0. For this calculation, information regarding the data set employed is given below in detail:

- TEİAŞ Turkey's electricity generation-consumption and loss statistics,
- Commonly prepared report under Turkey's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS) and Verified Carbon Standard (VCS) for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V2.0, Power plant efficiency figures are used

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<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

According to this publication;

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh

Combined Margin-CM (for other renewable than solar and wind); 0.5552²¹ tCO₂/MWh

5.2 Project Emissions

Project Emissions as per AMS-III.G (version 10)

Project emissions are calculated as follows as per AMS-III.G v10.

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y} \quad \text{(Equation-6)}$$

Where:

- PE_y** = Project emissions in year y (t CO₂/yr)
- PE_{Power,y}** = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)
- PE_{flare,y}** = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO₂e)
- PE_{process,y}** = Emissions from the landfill gas upgrading process in the year y (t CO₂e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

Determination of project emissions from electricity consumption (PE_{Power,y})

In accordance with paragraph 16 of AMS-III.G, “Project emissions from electricity consumption are determined as per the procedures described in the “Tool to calculate baseline, project and/or leakage emissions from electricity consumption”, “AMS-I.D.: Grid connected renewable electricity generation”. For project emissions from fossil fuel consumption, the “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” shall be used”.

According to Paragraph 16 of Tool 05, project emissions from the consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y}) \quad \text{(Equation-7)}$$

²¹ <https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

Where:

$PE_{EC,y}$	=	Project emissions from electricity consumption in year y (t CO ₂ / yr)
$EC_{PJ,j,y}$	=	Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
$EF_{j,y}$	=	Emission factor for electricity generation for source j in year y (t CO ₂ /MWh)
$TDL_{j,y}$	=	Average technical transmission and distribution losses for providing electricity to source j in year y
j	=	Sources of electricity consumption in the project

However, since the project activity involves electricity generation, it is expected that the plant will generate more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation will be claimed under AMS-I.D. v18.0 and energy consumption will be subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} will be considered only if there is energy consumption without energy generation in monthly basis. Therefore, $PE_{power,y}$ equals to 0.

Determination of project emissions from flare ($PE_{flare,y}$)

In the project activity there is no flare (there is a gas holder to retain extra gas) and no emission sourced from this. Therefore, $PE_{flare,y}$ equals to 0.

Determination of project emissions from landfill gas upgrading ($PE_{process,y}$)

In the project activity there is no landfill gas upgrading and no emission sourced from this. Therefore, $PE_{process,y}$ equals to 0.

Project Emissions as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO₂ emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”. Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed project activity, PE_y as per AMS-I.D., version 18.0., shall be accounted as $PE_y = PE_{flare,y}$. $PE_{flare,y}$ is taken as 0 as time of the validation since there is no flare unit in the project activity.

Therefore, the project emissions of the proposed project activity is 0.

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y} = 0$$

5.3 Leakage

According to paragraph 20 of AMS-III.G leakage must be considered if the methane recovery technology is transferred from another activity. As this project does not involve transfer of any equipment from another activity therefore the leakage has been considered zero.

Following paragraph 42 of AMS-I.D general guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues. Biomass residues are defined as per CDM glossary of Terms version 09.1 as “non-fossilized and biodegradable organic material originating from plants, animals and micro-organisms which is a by-product, residue or waste stream from agriculture, forestry and related industries”. This project does not pertain to the use of biomass residues. Therefore, no leakage effects need to be accounted for under the AMS-I.D and Leakage emission from the project activity is 0.

5.4 Estimated Net GHG Emission Reductions and Removals

The emission reduction achieved by the project activity is constituted by both the methane destruction and grid displacement components.

$$ER_y = BE_y - PE_y - LE_y \quad \text{(Equation-9)}$$

Where:

ER_y = Emission reductions in project year y (t CO₂)

BE_y = Baseline emissions in project year y (t CO₂)

PE_y = Project emissions in project year y (t CO₂)

LE_y = Leakage emissions in project year y (t CO₂)

As per AMS-III.G v10, paragraph 20, The actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y \quad \text{(Equation-10)}$$

Where:

$F_{CH4,PJ,y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

$$F_{CH4,PJ,y} = D_{CH4,y} \times w_{CH4,y} \times \sum_i LFG_{i,y} \quad \text{(Equation-11)}$$

Where:

$LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH4,y}$

$w_{CH4,y}$ = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$

D_{CH4} = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m³). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, $F_{CH4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \quad \text{(Equation-12)}$$

Where:

EG_y = Electricity generation in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH4} = NCV of methane (MJ/Nm³) use default value: 35.9 MJ/Nm³

EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation

- Default efficiency of 40 per cent

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December-2021	12,979	0	0	12,979
01-January-2022 – 31-December-2022	29,743	0	0	29,743
01-January-2023 – 31-December-2023	33,186	0	0	33,186
01-January-2024 – 31-December-2024	42,173	0	0	42,173
01-January-2025 – 31-December-2025	44,917	0	0	44,917
01-January-2026 – 31-December-2026	47,364	0	0	47,364
01-January-2027 – 31-December-2027	49,562	0	0	49,562
01-January-2028 – 30-June-2028	25,703	0	0	25,703
Total	285,627	0	0	285,627
Crediting Period Duration	7			
Average	40,803	0	0	40,803

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	Dimensionless
Description	Fraction of methane that would be oxidized in the top layer of the SWDS in the baseline
Source of data	Consistent with how oxidation is accounted for in the methodological tool “Emissions from solid waste disposal sites”
Value applied:	0.1
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per AMS-III.G v10
Purpose of Data	Baseline emission calculation
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC AR5 ²²
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC fifth Assessment Report (AR5). Shall be updated according to any future COP/MOP decisions.
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

²² <https://www.ipcc.ch/report/ar5/syr/>

Data / Parameter	η_{PJ}
Data unit	%
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	Feasibility study ²³
Value applied	85%
Justification of choice of data or description of measurement methods and procedures applied	c
Purpose of Data	Ex Ante Baseline emission calculation
Comments	-
Data / Parameter	ϕ_y
Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	<i>"Methodological Tool: Emissions from solid waste disposal sites" (Version 08.0)</i>
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	The project selects Option 1 :Use a default value to determine ϕ_y in the tool "Emissions from solid waste disposal sites" (version 08.0), and the project belongs to the situation of Application A: The project activity mitigates methane emissions from a specific existing SWDS, thus 0.75 is chosen for ϕ_y
Purpose of Data	Ex Ante baseline emission calculation
Comments	-

²³ Feasibility Report of Sinop Landfill Gas to Electricity Generation Facility

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per Tool04 - "Emissions from solid waste disposal sites"
Purpose of Data	Ex ante baseline emission calculation
Comments	-

Data / Parameter	DOC _{f,y}
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per Tool04-"Emissions from solid waste disposal sites"
Purpose of Data	Ex ante baseline emission calculation
Comments	-

Data / Parameter	MCF _y
Data unit	-
Description	Methane correction factor
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>

Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	The landfill is controlled, including the following activities: (i) cover material; (ii) mechanical compacting; (iii) levelling of the waste. Therefore, 1.0 for anaerobic managed solid waste disposal sites is recommended by “Emissions from solid waste disposal sites (Version 08.0).”
Purpose of Data	Baseline emission calculation
Comments	-

Data / Parameter	W_x																																								
Data unit	tons																																								
Description	Total amount of waste disposed in a SWDS in the year x																																								
Source of data	Feasibility study report (FSR) has been used for the waste amount. Waste compositions study of incoming waste to the landfill has been used for waste composition.																																								
Value applied	<u>Waste Amount:</u> <table border="1"> <thead> <tr> <th>Year</th><th>Total (tonnes)</th></tr> </thead> <tbody> <tr><td>2012</td><td>18,250</td></tr> <tr><td>2013</td><td>27,375</td></tr> <tr><td>2014</td><td>28,744</td></tr> <tr><td>2015</td><td>30,181</td></tr> <tr><td>2016</td><td>31,690</td></tr> <tr><td>2017</td><td>33,274</td></tr> <tr><td>2018</td><td>36,269</td></tr> <tr><td>2019</td><td>38,083</td></tr> <tr><td>2020</td><td>43,414</td></tr> <tr><td>2021</td><td>56,438</td></tr> <tr><td>2022</td><td>70,548</td></tr> <tr><td>2023</td><td>71,959</td></tr> <tr><td>2024</td><td>73,398</td></tr> <tr><td>2025</td><td>74,132</td></tr> <tr><td>2026</td><td>74,874</td></tr> <tr><td>2027</td><td>75,622</td></tr> <tr><td>2028</td><td>76,379</td></tr> <tr><td>2029</td><td>77,142</td></tr> <tr><td>2030</td><td>77,914</td></tr> </tbody> </table>	Year	Total (tonnes)	2012	18,250	2013	27,375	2014	28,744	2015	30,181	2016	31,690	2017	33,274	2018	36,269	2019	38,083	2020	43,414	2021	56,438	2022	70,548	2023	71,959	2024	73,398	2025	74,132	2026	74,874	2027	75,622	2028	76,379	2029	77,142	2030	77,914
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	2031	78,693
	2032	79,480
	2033	80,275
	2034	81,077
	2035	81,888
	2036	82,707
	2037	83,534
	2038	84,369
	2039	85,213
	2040	86,065
	<u>Waste Composition:</u>	
	Waste Type	Weight Fraction (%)
	Food	58.59
	Paper and Cardboard	7.06
	Wood	5.90
	Textile	3.66
	Garden, yard and park waste	0.00
	Inert materials (plastic, metal etc.)	24.79
	Total	100.00
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year x from FSR. For waste composition, sample amount of waste from different areas classed according to their wealth is taken and studied to determine waste composition by weight.	
Purpose of Data	Ex-ante calculation of baseline emissions	
Comments	-	

Data / Parameter	DOC _j	
Data unit	-	
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)	
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)	
Value applied	Waste type j	DOC _j (% wet waste)
	Wood and wood products	43
	Pulp, paper and cardboard (other than sludge)	40

	Food, food waste, beverages and tobacco (other than sludge)	15
	Textiles	24
	Garden, yard and park waste	20
	Glass, plastic, metal other inert waste	0
Justification of choice of data or description of measurement methods and procedures applied	Application A in the Tool04 has been chosen.	
Purpose of Data	Ex ante Baseline emission calculation	
Comments	-	

Data / Parameter	kj					
Data unit	1/yr					
Description	Decay rate for the waste type j					
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)					
Value applied	Waste type j		Boreal and Temperate (MAT≤20 °C)		Tropical (MAT>20 °C)	
			Dry (MAP/PET<1)	Wet (MAP/PET >1)	Dry (MAP< 1000mm)	Wet (MAP > 1000mm)
	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
		Wood, wood products and straw	0.02	0.03	0.025	0.035
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
	Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06	0.185	0.085	0.40

Justification of choice of data or description of measurement methods and procedures applied	Climatic conditions for Sinop where project site located in: MAT – mean annual temperature: 14.3 °C MAP – Mean annual precipitation: 689.3 mm PET – potential evapotranspiration: 150 mm Therefore, Boreal ($MAT \leq 20^{\circ}C$) and wet ($MAP/PET > 1$) conditions are observed in the project site. Sources: https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceleri-istatistik.aspx?k=undefined&m=SINOP https://www.mgm.gov.tr/arastirma/buharlasma.aspx?s=2020
Purpose of Data	Ex ante Baseline emission calculation
Comments	

Data / Parameter	ρ_{CH_4}
Data unit	t CH ₄ / m ³ CH ₄
Description	Density of methane at normal conditions
Source of data	‘Thermophysical properties of fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane’ by B.A. Younglove, J.F. Ely Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Value applied:	0.00067
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Purpose of Data	Calculation of baseline/project emissions
Comments	corrected to reference conditions

Data / Parameter	NCV _{CH₄}
------------------	-------------------------------

Data unit	MJ/Nm ³
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G. "Landfill methane recovery" (version 10.0).
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	Default Value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Calculated by Turkish Ministry of Energy and Natural Resources. ²⁴ Also, "Tool to calculate the emission factor for an electricity system (Version 07.0)" is applied.
Value applied	0.5552 tCO ₂ /MWh
Justification of choice of data or description of measurement methods and procedures applied	The data was obtained from Turkish Ministry of Energy and Natural Resources. Also, "Tool to calculate the emission factor for an electricity system (Version 07.0)" is applied.
Purpose of Data	For the calculation of the Baseline Emission

Data / Parameter	TDL _y
Data unit	%
Description	Average technical transmission and distribution losses

²⁴

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

Source of data	Obtained from https://webapi.teias.gov.tr/file/9248a331-e529-47ca-8c49-22f8705f5810?download
Value applied	10.0%
Justification of choice of data or description of measurement methods and procedures applied	-Most recent annual data is available for 2020 and losses from transmission and distribution are 1.7% and 8.3% respectively and total of 10%.
Purpose of Data	Calculation of project emissions
Comments	-

6.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh
Description	Quantity of net electricity supplied by the project activity to the grid in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	The quantity of electricity consumed by the project plant will be monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	-10,605 (until 2024) and 21,210 (after 2024) MWe for validation and will be monitored ex-post
Monitoring equipment	The quantity of electricity consumed by the project plant will be monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.
QA/QC procedures to be applied	<i>The electricity meters are used to measure the net electricity exported to the grid and imported from the grid and used for billing purposes. Therefore, the meters are sealed by TEİAŞ (Turkish Electricity Transmission CO) and/or through protocols where and when appropriate. The grid company is responsible for maintenance and calibration of the device. In accordance</i>

	with the “Regulation on “Measurement and Measuring Tools”, electricity meters have to be calibrated within a frequency of once per 10 years.
Purpose of data	Calculation of baseline emissions
Calculation method	This parameter is calculated by subtracting electricity imported from electricity supplied to the grid and formula is as follows: $EG_{PJ,y} = EG_{export,y} - EG_{import,y}$
Comments	-

Data / Parameter	EG _y
Data unit	MWh
Description	Quantity of gross electricity generation
Source of data	SCADA
Description of measurement methods and procedures to be applied	The quantity of generated electricity will be monitored by SCADA meter on generator with PLC system. Measurements will be monitored continuously and recorded by project staff monthly basis.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	Will be monitored ex-post.
Monitoring equipment	PLC
QA/QC procedures to be applied	The device will be calibrated periodically by equipment manufacturer
Purpose of data	Calculation of emission reduction
Calculation method	-
Comments	Since emission reduction will be calculated from electricity generation, this parameter shall be used.

Data / Parameter	EE _y
Data unit	%
Description	Energy conversion efficiency of the project equipment

Source of data	AMS-III.G. "Landfill methane recovery" (version 10.0).
Value applied	40%
Justification of choice of data or description of measurement methods and procedures applied	Default Value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$F_{CH_4,BL,R,y}$
Data unit	tCH ₄ /yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Source of data	Regulation in the country
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	As per regulation of the country there is no obligation to utilize or destroy methane
Purpose of Data	Calculation of baseline emissions
Comments	The parameter shall be used in "Case 2" of section 5.1

Data / Parameter	$p_{reg,y}$
Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns
Value applied	0

Justification of choice of data or description of measurement methods and procedures applied	As per regulation of the country there is no obligation to utilize or destroy methane
Purpose of Data	Calculation of baseline emissions
Comments	The parameter shall be used in “Case 2” of section 5.1

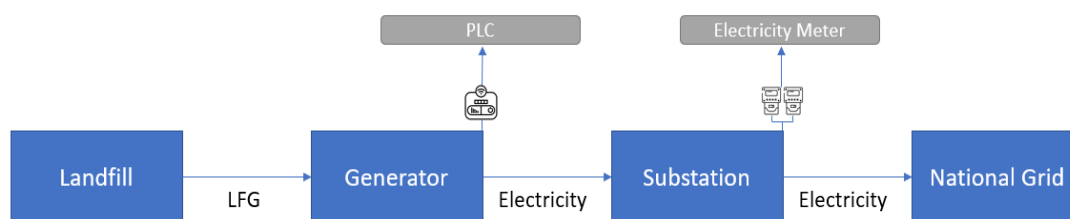
6.3 Monitoring Plan

Monitoring will be carried out following the procedures set by applicable methodology and all applicable tools of the above explained parameters. The monitoring consists of:

Responsibilities for monitoring

The project owner is responsible for the operation and maintenance of the landfill and the installed equipment. The project owner is also responsible for the administration of the data, setting up a VER team who will be responsible for monitoring all data required to estimate emission reductions.

Data and parameters to be monitored:



Monitoring Device	Monitored Parameter	Description
PLC	EG_y	Amount of gross electricity generated on alternator of genset using LFG by the project activity is continuously monitored by PLC device installed on alternator of the generator and continuously monitored in SCADA.
Electricity Meter	$EG_{PJ,y}$	Quantity of net electricity generation supplied by the project to the grid is continuously monitored by bi-directional electricity meter installed by outlet of onsite substation.

Registration of the monitored data

The Sinop LFG VER-Team will be responsible for quality assurance and quality control of the monitoring equipment. The data measured by the monitoring equipment will be stored and will be processed into a monitoring report, which will be submitted by the project owner. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team will regularly check the monitoring system for errors. In the case of errors, corrective actions will be undertaken by the VER-Team, or if required, by the supplier of the monitoring equipment.

QA/QC procedure

Strong quality assurance and quality control procedures will be taken to monitor the equipment and data collection. Equipment and facilities will be subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or in correct data reading, the data will be crosschecked with other parameters and data leading to the most conservative emission reductions will be considered for the calculation of the emission reductions.

Measuring of exported and imported electricity

The main meter that monitors electricity import and export is placed at the grid connection. Monthly protocols based on the readings that are confirmed by both TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed project activity. The electricity meters are bidirectional and will measure both the imported and exported electricity from and to the grid.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	EGPJ,y																									
Data unit	MWh																									
Description	Quantity of net electricity supplied by the project activity to the grid in year y																									
Value applied	Total Generation is 8,123 MWh <table><tr><th>Year</th><th>Net electricity supplied to the grid[MWh]</th></tr><tr><td>01 July 2021 – 31 December 2022</td><td>2,470.86</td></tr><tr><td>01 January 2022 – 31 December 2022</td><td>4,838.53</td></tr><tr><td>01 January 2023 – 28 February 2023</td><td>813.57</td></tr><tr><td>Total Sum 01-July-2021-28-February-2023</td><td>8,123</td></tr></table> Monitoring equipment (electricity meter) specifications: <table><tr><th>Equipment</th><th>Serial Number</th><th>Brand & Model</th><th>Accuracy Class</th><th>Date of Calibration</th></tr><tr><td>Main meter</td><td>80276622</td><td>Makel – C520KMT2556</td><td>C Class (0.5)</td><td>26-May-2021</td></tr><tr><td>Backup meter</td><td>80272124</td><td>Makel – C520KMT2556</td><td>C Class (0.5)</td><td>26-May-2021</td></tr></table>	Year	Net electricity supplied to the grid[MWh]	01 July 2021 – 31 December 2022	2,470.86	01 January 2022 – 31 December 2022	4,838.53	01 January 2023 – 28 February 2023	813.57	Total Sum 01-July-2021-28-February-2023	8,123	Equipment	Serial Number	Brand & Model	Accuracy Class	Date of Calibration	Main meter	80276622	Makel – C520KMT2556	C Class (0.5)	26-May-2021	Backup meter	80272124	Makel – C520KMT2556	C Class (0.5)	26-May-2021
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Backup meter	80272124	Makel – C520KMT2556	C Class (0.5)	26-May-2021																						
Comments	The data is monitored by the bi-directional electricity meter that installed at the outlet of the onsite substation with the accuracy of 0.5. Meter readings are read and recorded by YEDAS (Distribution Company) and EPIAS. Since the electricity is transferred via energy transmission line EPIAS deduct this amount and EPIAS system shows invoice-based reading by excluding transmission losses. EPIAS data will be the main data and YEDAS readings will be used for cross-check.																									

Data / Parameter	EG _y																		
Data unit	MWh																		
Description	Quantity of gross electricity generation																		
Value Applied	Total Generation 8,653 <table><thead><tr><th>Year</th><th>Electricity generated [MWh]</th></tr></thead><tbody><tr><td>01 July 2021 – 31 December 2022</td><td>2,622.44</td></tr><tr><td>01 January 2022 – 31 December 2022</td><td>5,158.16</td></tr><tr><td>01 January 2023 – 28 February 2023</td><td>872.04</td></tr><tr><td>Total Sum 01-July-2021-28-February-2023</td><td>8,653</td></tr></tbody></table> Monitoring equipment (electricity meter) specifications: <table><thead><tr><th>Equipment</th><th>Brand & Model</th><th>Rated Voltage</th><th>Supply Voltage</th></tr></thead><tbody><tr><td>PLC</td><td>Bender – FP200</td><td>50 V</td><td>24 V</td></tr></tbody></table>	Year	Electricity generated [MWh]	01 July 2021 – 31 December 2022	2,622.44	01 January 2022 – 31 December 2022	5,158.16	01 January 2023 – 28 February 2023	872.04	Total Sum 01-July-2021-28-February-2023	8,653	Equipment	Brand & Model	Rated Voltage	Supply Voltage	PLC	Bender – FP200	50 V	24 V
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Equipment	Brand & Model	Rated Voltage	Supply Voltage																
PLC	Bender – FP200	50 V	24 V																
Comments																			

Data / Parameter	EE _y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Value Applied	40%
Comments	<i>Default value in AMS-III.G v 10 has been used.</i>

Data / Parameter	F _{CH4,BL,R,y}
Data unit	tCH ₄ /yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Value applied	0
Comments	Since there is no regulation Case 1 is valid during monitoring period.

Data / Parameter	p _{reg,y}
------------------	--------------------

Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Value applied	0
Comments	Since there is no regulation Case 1 is valid during monitoring period.

7.2 Baseline Emissions

Baseline emission reductions are calculated in accordance with AMS-III.G (Version 10.0) and AMS-I.D. (Version 18.0). The baseline emissions are calculated by the following equation:

$$BE_y = BE_{CH_4,y} + BE_{electricity,y} \quad \text{(Equation-13)}$$

Where;

$$BE_y = \text{Baseline emissions in year } y \text{ (tCO}_2\text{e/y)}$$

$$BE_{CH_4,y} = \text{Baseline emissions of methane from the SWDS in year } y \text{ (tCO}_2\text{e/y)}$$

$$BE_{electricity,y} = \text{Baseline emissions associated with electricity displacement } y \text{ (tCO}_2\text{e/y)}$$

Determination of Baseline Emissions from AMS-III.G. ($BE_{CH_4,y}$)

As per AMS-III.G, the actual emission reduction achieved for Type III component of the project (which means not including emission reduction from Type I component, i.e. LFG power generation) during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y \quad \text{(Equation-14)}$$

Where:

$$F_{CH_4,PJ,y} = \text{Methane captured and destroyed/gainfully used by the project activity in the year } y \text{ (tCH}_4\text{)}$$

Determination of $F_{CH_4,BL,y}$

As per ACM0001 v19.0, 4 cases given in the table below has been evaluated.

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Since there is no existing LFG capture system prior to project activity and there is no requirement to destroy methane Case 1 is the project activity case. Therefore, $F_{CH4,BL,y}$ equals to 0.

$$F_{CH4,PJ,y} = D_{CH4,y} \times w_{CH4,y} \times \sum_i LFG_{i,y} \quad \text{(Equation-15)}$$

Where:

- $LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m^3LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH4,y}$
- $w_{CH4,y}$ = Methane content in landfill gas in year y (volume fraction, m^3CH_4/m^3LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$
- D_{CH4} = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/ m^3). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, $F_{CH4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \quad \text{(Equation-16)}$$

Where:

- EG_y = Electricity generation in year y (MWh)
- 3600 = Conversion factor (1 MWh = 3600 MJ)
- NCV_{CH4} = NCV of methane (MJ/ Nm^3) use default value: 35.9 MJ/ Nm^3
- EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
- Default efficiency of 40 per cent

There is no mandatory regulatory requirement on landfill gas destruction/combustion that has been systematically enforced in Turkey which means there is no contractual requirement on methane destruction that the project owner is bound to. Therefore, $F_{CH4,BL,y}$ is 0.

According to equations above EG_y , $F_{CH4,PJ,y}$ and finally $BE_{CH4,y}$ in this monitoring is given in the table below:

Months	EG_y (MWh)	$F_{CH4,PJ,y}$ (tCH ₄)	$F_{CH4,BL,y}$ (tCH ₄)	BE_{CH4} (tCO _{2e})
Jul-21	272.75	45.81	0	1,154
Aug-21	498.79	83.78	0	2,111
Sep-21	522.67	87.79	0	2,212
Oct-21	470.92	79.10	0	1,993
Nov-21	436.93	73.39	0	1,849
Dec-21	420.39	70.61	0	1,779
Sum 2021	2,622.44	440.48	0	11,100
Jan-22	441.82	74.21	0	1,870
Feb-22	422.01	70.88	0	1,786
Mar-22	450.74	75.71	0	1,908
Apr-22	428.12	71.91	0	1,812
May-22	418.30	70.26	0	1,771
Jun-22	446.22	74.95	0	1,889
Jul-22	506.04	85.00	0	2,142
Aug-22	457.29	76.81	0	1,936
Sep-22	415.79	69.84	0	1,760
Oct-22	371.22	62.35	0	1,571
Nov-22	364.69	61.26	0	1,544
Dec-22	435.91	73.22	0	1,845
Sum 2022	5,158.16	866.40	0	21,833
Jan-23	430.34	72.28	0	1,822
Feb-23	441.70	74.19	0	1,870
Sum 2023	872.04	146.47	0	3,691
Total Sum 01-July-2021 - 28- February-2023	8,653	1,453	0	36,624

Determination of Baseline Emissions from AMS-III.G. ($BE_{\text{electricity},y}$)

Project activity utilize LFG to generate renewable energy and generated electricity is supplied to the national grid.

As per AMS-I.D. (version 18.0) baseline emission calculated as follows:

$$BE_{y,\text{electricity}} = EG_{PJ,y} + EF_{\text{grid},y} \quad \text{(Equation-17)}$$

Where;

$BE_{\text{electricity},y}$	=	Baseline emissions associated with electricity displacement y (tCO _{2e} /y)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{\text{grid},y}$	=	Combined margin CO ₂ emission factor of grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO ₂ /MWh)

$$EG_{PJ} = EG_{PJ,\text{facility},y} \quad \text{(Equation-18)}$$

Where;

$EG_{PJ,\text{facility},y}$	=	Quantity of net electricity generation that is produced by the project plant to the grid in year y (MWh)
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Emission factors of the Turkish National Grid is as follows by the publications of Ministry of Energy and Natural Sources²⁵:

According to this publication;

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh

Combined Margin-CM (for other renewable than solar and wind); 0.5552 tCO₂/MWh

Therefore, baseline emissions related to electricity generations is as follows

²⁵ <https://enerji.gov.tr//Media/Dizin/EVCED/tr/ÇevreVeİklim/İklimDeğişikliği/TUESEmisyonFktr/Belgeler/Bform2020.pdf>

Months	EG _{PJ, facility, y} (MWh)	BE _{electricity, y} (tCO _{2e})
Jul-21	255.30	142
Aug-21	470.37	261
Sep-21	493.60	274
Oct-21	444.01	247
Nov-21	411.84	229
Dec-21	395.74	220
Sum 2021	2,470.86	1,371
Jan-22	416.09	231
Feb-22	397.84	221
Mar-22	424.57	236
Apr-22	403.15	224
May-22	393.41	218
Jun-22	420.56	233
Jul-22	477.74	265
Aug-22	430.64	239
Sep-22	386.00	214
Oct-22	343.71	191
Nov-22	338.45	188
Dec-22	406.36	226
Sum 2022	4,838.53	2,686
Jan-23	401.25	223
Feb-23	412.32	229
Sum 2023	813.57	451
Total Sum 01-July-2021 - 28- February-2023	8,123	4,508

As a result of these, sum of baseline emission BE_y is as follows:

Months	BE _{electricity, y} (tCO _{2e})	BE _{CH₄} (tCO _{2e})	BE _y (tCO _{2e})
Jul-21	142	1,154	1,296
Aug-21	261	2,111	2,372
Sep-21	274	2,212	2,486
Oct-21	247	1,993	2,240
Nov-21	229	1,849	2,078
Dec-21	220	1,779	1,999
Sum 2021	1,371	11,100	12,471
Jan-22	231	1,870	2,101
Feb-22	221	1,786	2,007
Mar-22	236	1,908	2,144
Apr-22	224	1,812	2,036
May-22	218	1,771	1,989

<i>Jun-22</i>	233	1,889	2,122
<i>Jul-22</i>	265	2,142	2,407
<i>Aug-22</i>	239	1,936	2,175
<i>Sep-22</i>	214	1,760	1,974
<i>Oct-22</i>	191	1,571	1,762
<i>Nov-22</i>	188	1,544	1,732
<i>Dec-22</i>	226	1,845	2,071
Sum 2022	2,686	21,833	24,519
<i>Jan-23</i>	223	1,822	2,044
<i>Feb-23</i>	229	1,870	2,099
Sum 2023	451	3,691	4,142
Total Sum 01-July-2021 - 28- February-2023	4,508	36,624	41,132

7.3 Project Emissions

Project Emissions as per AMS-III.G.

Project emissions are calculated as follows as per AMS-III.G v10.

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y} \quad \text{(Equation-19)}$$

Where:

- PE_y = Project emissions in year y (t CO₂/yr)
- $PE_{Power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)
- $PE_{flare,y}$ = Emissions from flaring or combustion of the landfill gas stream in the year y (t CO₂e)
- $PE_{process,y}$ = Emissions from the landfill gas upgrading process in the year y (t CO₂e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

Since the project activity involves electricity generation, it is expected that the plant will generate more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation will be claimed under AMS-I.D. v18.0 and energy consumption will be subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} is zero.

In the project site, there is gas holder to retain extra gas, there is no flare unit. Hence PE_{flare} is taken as zero.

The project does not involve landfill gas upgrading process, hence PE_{process} is also zero.

Project Emissions as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO_2 emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the “Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion”. Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed project activity, PE_y as per AMS-I.D., version 18.0., shall be accounted as 0.

In conclusion, PE_y for the project activity is 0.

7.4 Leakage

No leakage effects are accounted for under AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).

7.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-July-2021 – 31-December 2021	12,471	0	0	12,471
01-January-2022 – 31-December 2022	24,519	0	0	24,519

01-January-2023 – 28 February 2023	4,142	0	0	4,142
Total	41,132	0	0	41,132

Comparison of ex-ante emission reductions and actual emission reduction is given in the table below.

Vintage	Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
01-July-2021 – 31-December 2021	12,979 tCO ₂ e.	12,471 tCO ₂ e	-4%	Ex-post calculations are slightly higher than ex-ante emission reductions and they are quite close each other.
01-January-2022 - 31-December 2022	29,743 tCO ₂ e.	24,519 tCO ₂ e	-21%	Ex-ante calculations are slightly higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions, so the difference is acceptable such project.
01-January-2023 - 28-February 2022	5,364 tCO ₂ e.	4,142 tCO ₂ e	-30%	Ex-ante calculations are slightly higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions, so the difference is acceptable such project.
01-July-2021 – 28-February 2022	48,086 tCO ₂ e.	41,132 tCO ₂ e.	-17%	Ex-ante calculations are slightly higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions. By considering this, the difference between ex-ante and ex-post calculations are quite close.

APPENDIX 1:

Amount of Emission Reductions per methodology per month:

SINOP EMISSION REDUCTION AMOUNT (01-July-2021 - 28-February-2023)			
Months	Net Emission Reduction by electricity (as per AMS-I.D.) (tCO ₂ e)	Net Emission Reduction by LFG capture (as per AMS-III.G.) (tCO ₂ e)	Total Net Emission Reduction (tCO ₂ e)
<i>Jul-21</i>	142	1,154	1,296
<i>Aug-21</i>	261	2,111	2,372
<i>Sep-21</i>	274	2,212	2,486
<i>Oct-21</i>	247	1,993	2,240
<i>Nov-21</i>	229	1,849	2,078
<i>Dec-21</i>	220	1,779	1,999
Sum 2021	1,371	11,100	12,471
<i>Jan-22</i>	231	1,870	2,101
<i>Feb-22</i>	221	1,786	2,007
<i>Mar-22</i>	236	1,908	2,144
<i>Apr-22</i>	224	1,812	2,036
<i>May-22</i>	218	1,771	1,989
<i>Jun-22</i>	233	1,889	2,122
<i>Jul-22</i>	265	2,142	2,407
<i>Aug-22</i>	239	1,936	2,175
<i>Sep-22</i>	214	1,760	1,974
<i>Oct-22</i>	191	1,571	1,762
<i>Nov-22</i>	188	1,544	1,732
<i>Dec-22</i>	226	1,845	2,071
Sum 2022	2,686	21,833	24,519
<i>Jan-23</i>	223	1,822	2,044
<i>Feb-23</i>	229	1,870	2,099
Sum 2023	451	3,691	4,142
Total Sum 01-July-2021 - 28-February-2023	4,508	36,624	41,132